

Power Generation Costs and Impacts on Electric Bills



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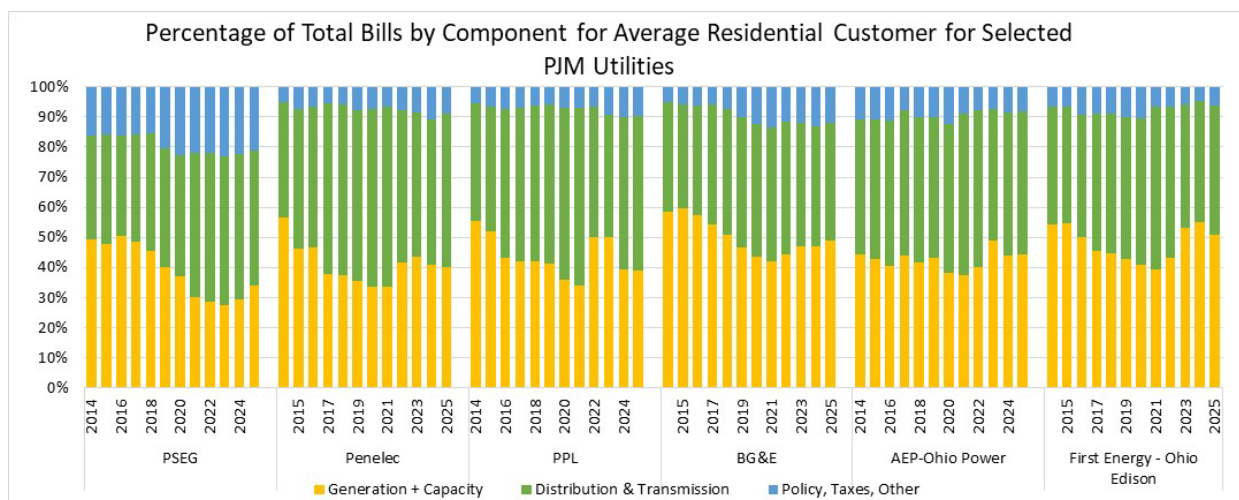
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Executive Summary of Rate Study Results

The Electric Power Supply Association (EPSA) retained Energy Tariff Experts, LLC (ETE) to conduct a study of average residential retail electric bills over the past decade to evaluate the costs of generation from the PJM Interconnection market along with other costs such as distribution, transmission, and public policy programs that make up the total bill. ETE found that the generation component of residential customer bills has fluctuated over the study period, averaging approximately 45% of the total bill. Rising costs in most utility service areas studied can be attributed to increasing transmission charges and, particularly in New Jersey and Maryland, public policy related charges have grown at rates exceeding inflation.

In 2025, the residential retail cost of generation as a percentage of customer bills has not increased materially and is consistent with historical averages.

This is illustrated in the graphic of representative PJM utilities below.^{1,2}

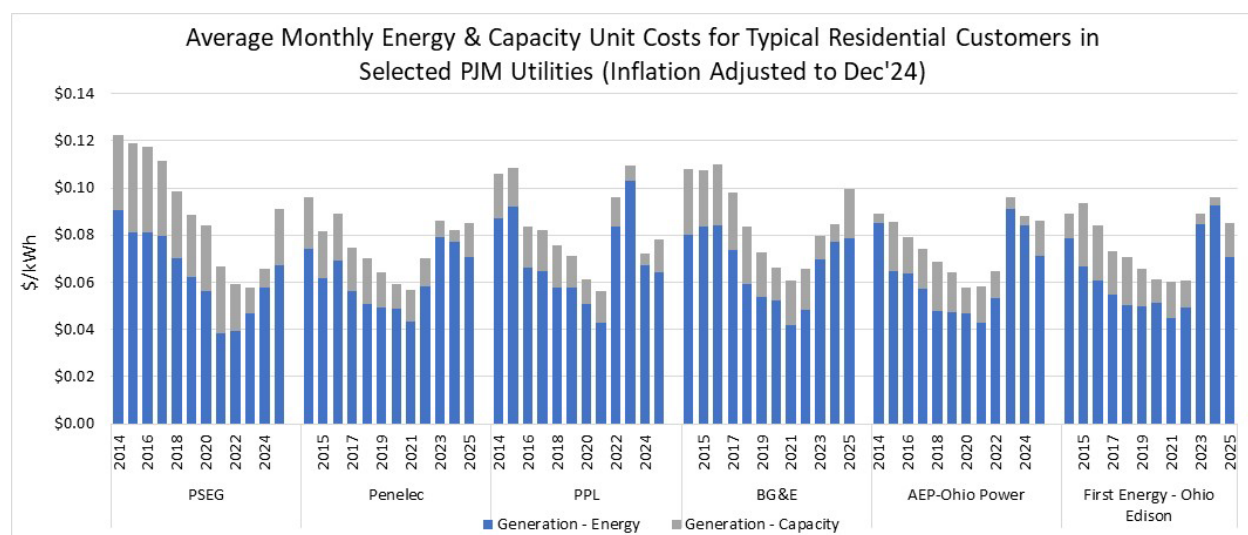


The generation component of the bill consists of both energy and capacity since these markets work together to ensure reliable energy supply. Throughout the study period, costs for energy and capacity have varied, but often in divergent directions.

¹ ETE's data sources include a continuous replication of effective utility rates and riders from January 2014 to present obtained from utility websites or state Commission file rooms. Average customer bills were calculated using average monthly customer usage which was obtained from a combination of rate case workpapers, competitive supplier shopping data, and workpapers associated with standard offer procurement filings. Isolation of RPS compliance costs was conducted using a combination of data contained in state level RPS compliance reports or calculated mathematically using the annual percentage RPS requirements by REC type and prevailing prices in RPS broker surveys.

² Utilities chosen for display in Executive Summary graphs are either the largest in their state by residential customer count or in the case of Penelec, span a wide geography.

ETE's study relied upon publicly available utility data such as standard offer default supply rates for residential customers in each utility which generally consist of laddered forward auctions to procure generation and capacity. When capacity is embedded in the generation price, ETE determined the cost of capacity using average customer contribution to peak load (aka capacity tags) and PJM Reliability Pricing Model (RPM) final auction prices to isolate the capacity component in generation charges. Depending upon the utility, other embedded components such as Renewable Portfolio Standard (RPS) compliance costs and transmission charges were also reconstituted to calculate the portion of generation charges solely attributable to generation energy and capacity. The resulting annual average unit costs for generation energy and capacity are shown in the table below.^{3,4}



The graphic above shows that on an inflation adjusted basis, 2025 is not an outlier in terms of generation energy and capacity costs relative to prior years in the study period.

The differences in calculated monthly generation costs across utilities, in total and on a unit cost basis, are attributable to monthly customer usage patterns for the average customer, geographic differences in PJM prices, and regulatory lag in the standard offer pricing mechanisms for default generation service. Standard offer prices for June 2025 onward were estimated based on known default service procurements, calculated capacity charges, and replication of standard offer workpapers where applicable.

In conclusion, in calendar year 2025, total generation costs as a percentage of the total bill and in inflation adjusted unit cost terms are consistent with historical averages for residential customers. In

³ ETE's methodology centered on calculating the bill for an average usage customer. Average usage was found by dividing the total monthly residential MWh sales by the residential customer count for the most recent available time period. There are stark differences among the utilities in PJM with PPL and PSEG customers using an annual average of 11,325 kWh and 6,478 kWh, respectively. As a result, utilities with lower unit costs may still have higher average bills in dollar terms based on typical customer usage patterns.

⁴ New Jersey generation unit costs do not include Sales and Use Tax

June 2025, capacity prices will rise, but energy prices are currently low relative to recent years and the combined total of energy and capacity costs is consistent with prior years.

ABOUT THE AUTHOR

James Bride is the President of ETE, a Boston based consulting firm focused on utility rates and regulatory matters. James has 18 years of utility sector experience, has appeared as an expert witness in dozens of state commission proceedings throughout the US, and has an MBA from Cornell University.

Understanding Your Electric Bill: New Jersey

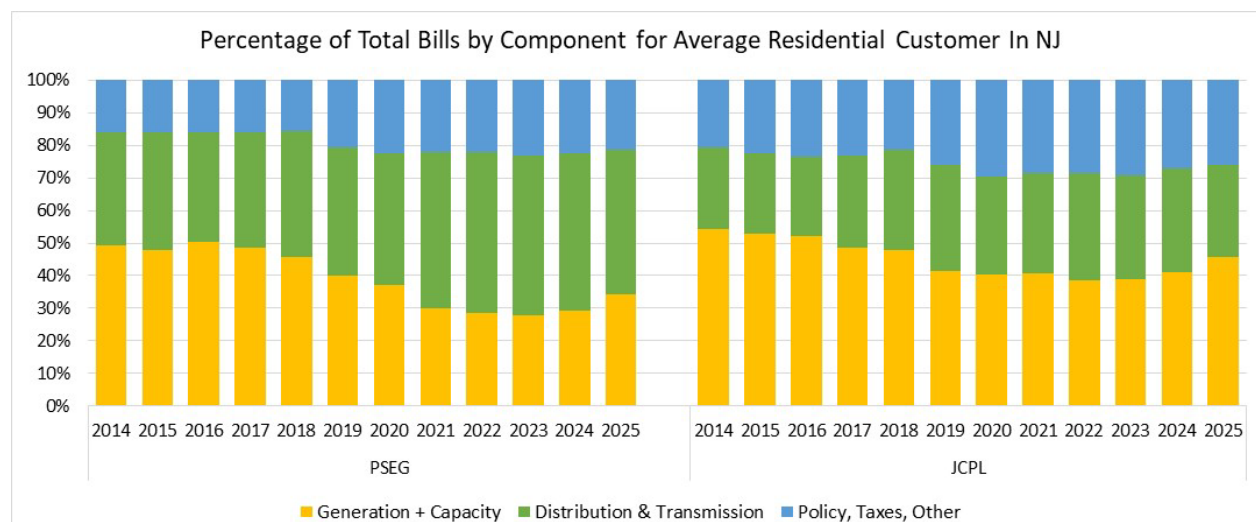
Energy Tariff Experts, LLC (ETE) conducted an analysis of electric utility bills for the average residential customer in the PJM region order to better understand how costs for different segments of the electric grid have contributed to the total bill over the past decade.

Analysis and Methods

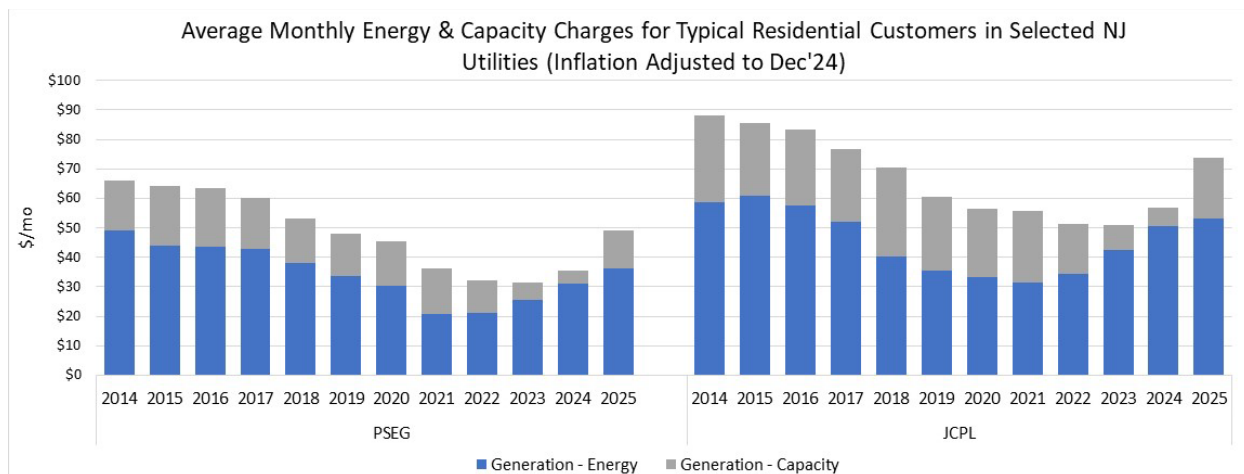
ETE assembled multiple primary sources to replicate a typical residential invoice by component. The electric bill for an average residential customer was replicated based on effective electric tariffs and other relevant data. The end product is additive of the total residential electric costs, by component over the past decade to show growth trends in total bills and underlying cost drivers. To learn more, see the methodology below.

SUMMARY OF FINDINGS

In its analysis of New Jersey residential electric bills, ETE found that the generation component of residential customer bills has fluctuated over the study period, averaging approximately 42% of the total bill. In 2025, the residential retail cost of generation as a percentage of customer bills has not increased materially and is consistent with historical averages. Rising transmission charges, particularly in PSE&G, and increasing public policy related charges are responsible for the majority of bill increases for residential consumers over the study period. The chart below shows how these impacts change by specific utility:

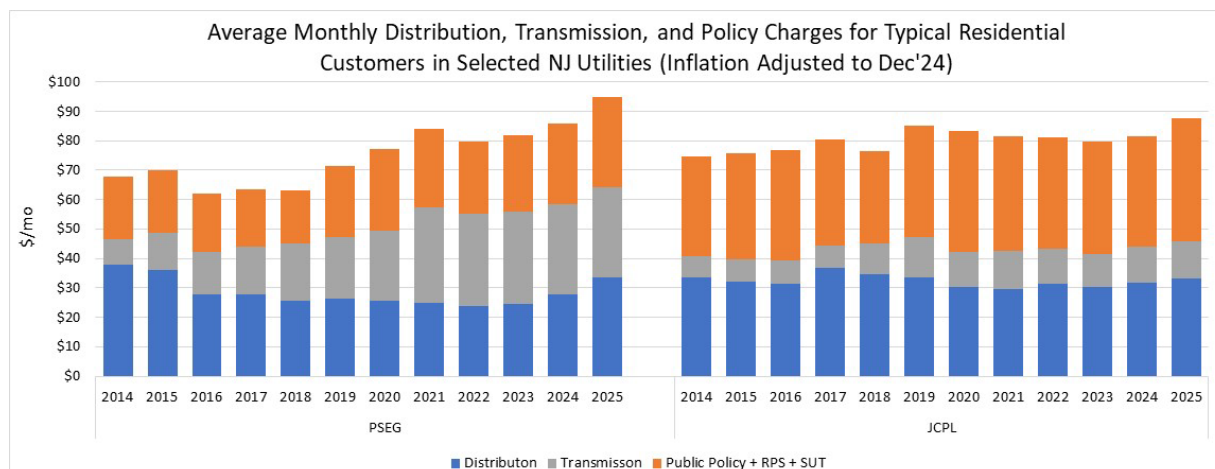


The generation component of the bill consists of both energy and capacity since these markets work together to ensure reliable energy supply. Throughout the study period, costs for energy and capacity have varied, but often in divergent directions.

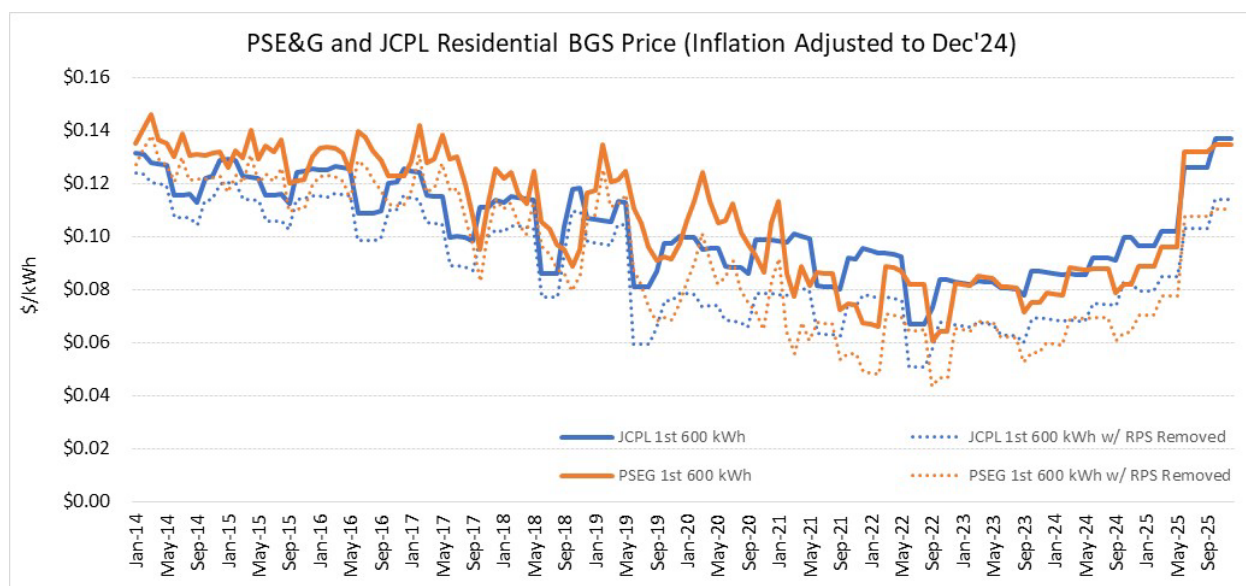


On an inflation adjusted basis, 2025 is not an outlier in terms of generation energy and capacity costs relative to prior years in the study period. Basic Generation Service (BGS) prices for June 2025 onward were obtained from the most recent utility compliance filings to the New Jersey Board of Public Utilities (BPU) in the BGS docket. The capacity component of BGS rates were calculated using the average residential generation capacity obligation (tag) and the final zonal capacity prices resulting from the completion of the third incremental auction. The differences in monthly generation costs across New Jersey utilities can be attributable to average monthly customer usage and differences in PJM zonal capacity prices in some years.

For New Jerseyans, total generation costs as a percentage of their total bill and in inflation adjusted dollar terms have not increased materially relative to historical averages for residential customers. In June 2025, capacity prices will rise, but the combined inflation adjusted total of energy and capacity costs is consistent with prior years. Other components of the residential bill such as Transmission, Distribution, and Public Policy driven charges have also changed in the past decade as shown in the graph below. It is noteworthy that PSE&G shows lower average monthly billed generation charges relative to JCP&L in the above chart due to significantly lower average customer usage volumes, but has higher average monthly bills for other components with transmission charges being significantly in excess of those for JCP&L.



The Price to Compare (PTC) is the default price for generation supply and is applicable to customers who don't select a competitive supplier or participate in a Community Choice Aggregation (CCA). It updates annually with a summer and winter season and is generally a good proxy for generation costs for residential customers. In unit cost terms, the energy and capacity components of BGS are similar to many years in the past decade on an inflation adjusted basis.⁵ The dashed line shows the impact of backing out Renewable Portfolio Standard (RPS) charges which are embedded in BGS prices. Generation costs shown are inclusive of monthly reconciliation charges and Sales and Use Tax (SUT). The second price tier for usage over 600 kWh in the summer months has been omitted from the graph to improve readability.



⁵ JCP&L has always shown transmission charges separately in its PTC, but PSE&G didn't specifically enumerate transmission charges separately in its tariff until February 2021. ETE estimated by generation component of PSE&G's BGS prices by subtracting the volumetric transmission cost component indicated in the Company Specific Addendum Workpapers for each BGS year.

Methodology

Each utility specific analysis begins with the published tariff rate books approved by the New Jersey Board of Public Utilities (BPU). ETE assembled a continuous dataset of approved rates using its internal tariff library, utility websites, the BGS auction data rooms, and the BPU docket system. The rate buildup files include all rate components and relevant meta-data such as tariff sheets and revision numbers.

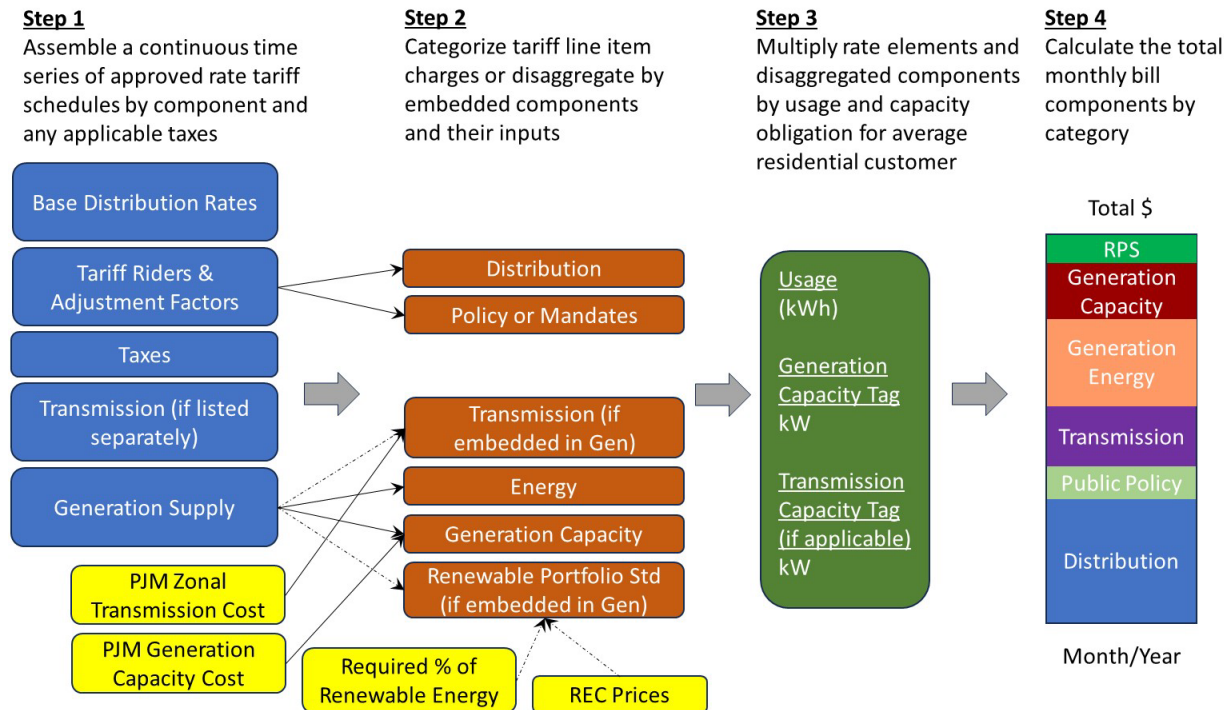
In some cases, there are multiple embedded costs included in a tariff rate item. This is true for BGS generation supply charges where energy, capacity, Renewable Portfolio Standard (RPS) charges are typically comingled. Historically in PSE&G prior to 2021, Transmission was also included in the default generation supply charge. ETE unbundled the generation charge into its components using the methods described below:

- **RPS Charges**
 - ETE obtained the historical annual percentages of energy required in each category of renewables and reported average Renewable Energy Credit (REC) prices. Multiplication of the required percentages and prevailing REC prices provides the unit cost in \$/MWh of RPS compliance. ETE also obtained RPS costs from the BGS Company Specific Addendums and utilized those when available.
- **Generation Capacity**
 - ETE assembled a time series of zonal capacity prices from the PJM Reliability Pricing Model (RPM) capacity auctions along with scaling factors such as line losses, forecast pool requirements, and zonal scaling factors. The final zonal capacity prices resulting from last incremental auction, if applicable, were used in this analysis.
 - Residential class average generation capacity obligation in kW (aka, “capacity tags”) were used to calculate the average residential customer cost responsibility for capacity
 - The scaled capacity prices were adjusted to daily \$/kW-day values and multiplied by the average residential capacity tag and number of days in a month to determine the total cost of generation capacity.
 - Volumetric equivalent capacity prices reported in the BGS Addendum filings were utilized in some years in PSEG.
- **Transmission (if applicable)**
 - In cases where PJM transmission charges were embedded in supply rates, ETE assembled a time series of historical zonal Network Integration Transmission Service (NITS) charges and converted them to \$/kW-day.
 - ETE obtained the average residential transmission capacity obligation values in kW (aka Transmission tags)
 - The transmission cost responsibility for the average residential customer was determined by multiplying the average residential capacity tags scaled for line losses by the effective daily NITS rate and the days in a month.

- **Generation Energy**
 - The total cost of the energy component of the generation charge was found deductively. The monthly cost of RPS compliance, generation capacity, and transmission (if applicable) were subtracted from the total cost of default generation supply and the remainder was ascribed to the energy component.
- **Public Policy Charges⁶**
 - In PSEG, the following Riders were designated as public policy related: Societal Benefits Charge; Solar Pilot Recovery Charge; Zero Emission Certificate Charges; and Green Programs Recovery Charge.
 - In JCP&L, the Non-Utility Generation Charge; Societal Benefits Charge; Demand Side Factor; Universal Service Fund Recovery; RGGI Recovery; and Zero Emission Certificate Charges were labeled as public policy.
- **Taxes**
 - In New Jersey, residential electric bills are subject to Sales & Use Tax (SUT). SUT costs embedded in tariff charges were calculated and separately identified.

⁶ The Riders to recover the costs of Zero Emission Certificates in all New Jersey utilities are expected to sunset in June 2025 or shortly thereafter and were removed from calculated 2025 values from June onward.

Figure 1: Illustration of Steps to Replicate Monthly Average Residential Electric Bill



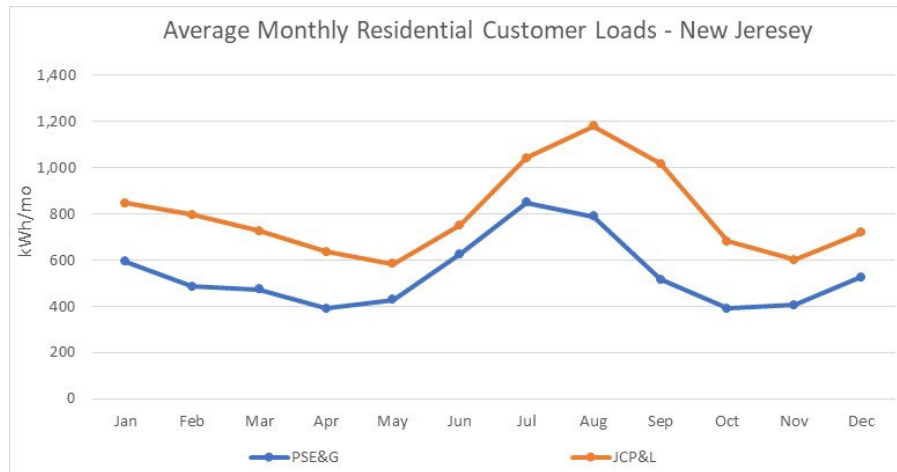
In New Jersey, residential bills were calculated for PSE&G and JCPL.

In addition to calculating bills in nominal dollars, ETE also calculated an inflation adjusted residential electric bill using the Consumer Price Index for All Urban Consumers (CPI-U) with a reference point of December 2024.

Definition of Average Residential Customer

In New Jersey, ETE utilized a combination of the data made available to retail energy suppliers participating in the default generation supply service auctions and workpapers from utility BPU filings in distribution proceedings to determine usage for the average residential customer. This data includes total residential customer counts, monthly consumption, and generation capacity obligations for the class. A minimum of 24 months of data was used to develop an annual profile of average residential monthly usage. The average residential consumption patterns by New Jersey utility are shown in the graph below.

Figure 2



This step is important because this analysis is focused on total customer bills. Usage trends vary by utility service area due to climate, average dwelling sizes, and home characteristics such as electric heat. As a result, comparing total bills can be a more relevant assessment of consumer costs than comparing unit costs in \$/kWh, by bill component.

Understanding Your Electric Bill: Pennsylvania

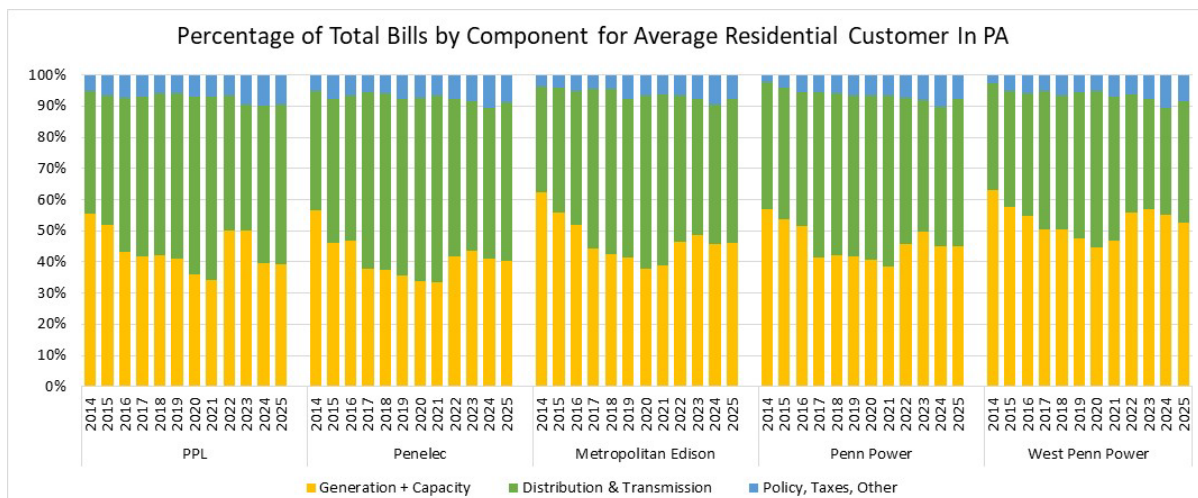
Energy Tariff Experts, LLC (ETE) conducted an analysis of electric utility bills for the average residential customer in the PJM region order to better understand how costs for different segments of the electric grid have contributed to the total bill over the past decade.

Analysis and Methods

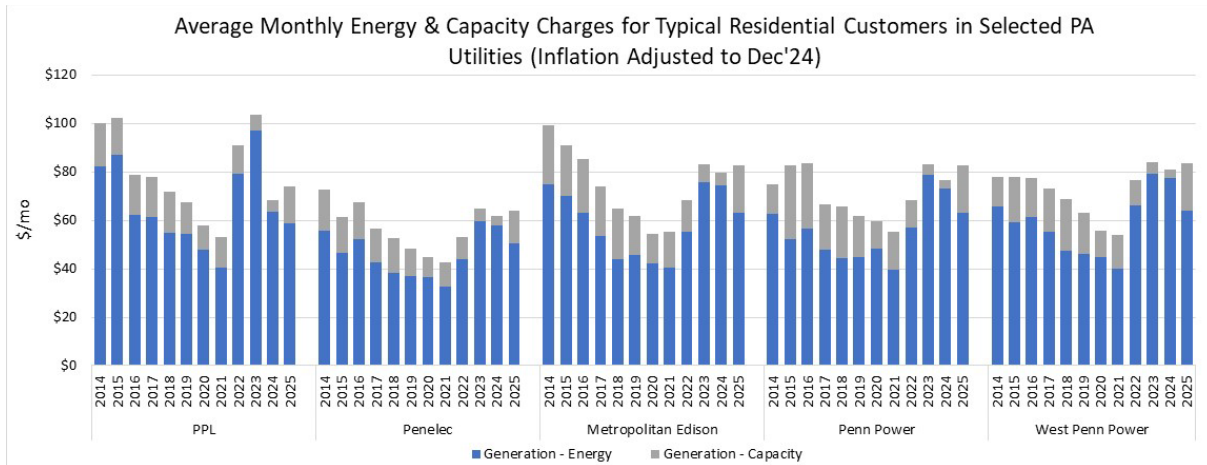
ETE assembled multiple primary sources to replicate a typical residential invoice by component. The electric bill for an average residential customer was replicated based on effective electric tariffs and other relevant data. The end product is additive of the total residential electric costs, by component over the past decade to show growth trends in total bills and underlying cost drivers. To learn more, see the methodology below.

SUMMARY OF FINDINGS

In its analysis of Pennsylvania residential electric bills, ETE found that the generation component of residential customer bills has fluctuated over the study period, averaging approximately 46% of the total bill. In 2025, the residential retail cost of generation as a percentage of customer bills has not increased materially above 2024 levels and is consistent with historical averages. The data indicates that transmission charges in several Pennsylvania utilities have grown substantially. The chart below shows how these impacts change by specific utility:

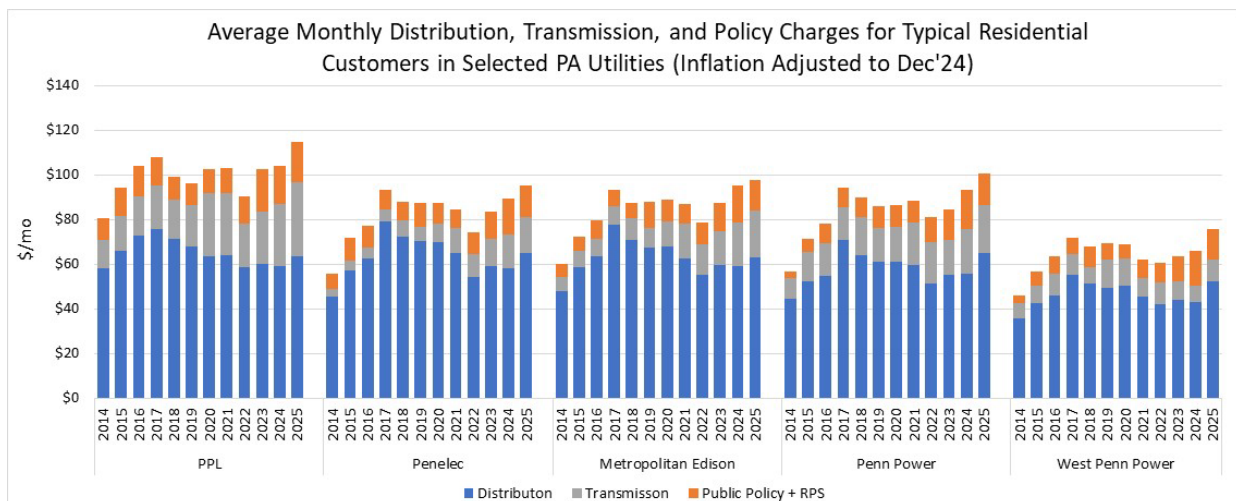


The generation component of the bill consists of both energy and capacity since these markets work together to ensure reliable energy supply. Throughout the study period, costs for energy and capacity have varied, but often in divergent directions.

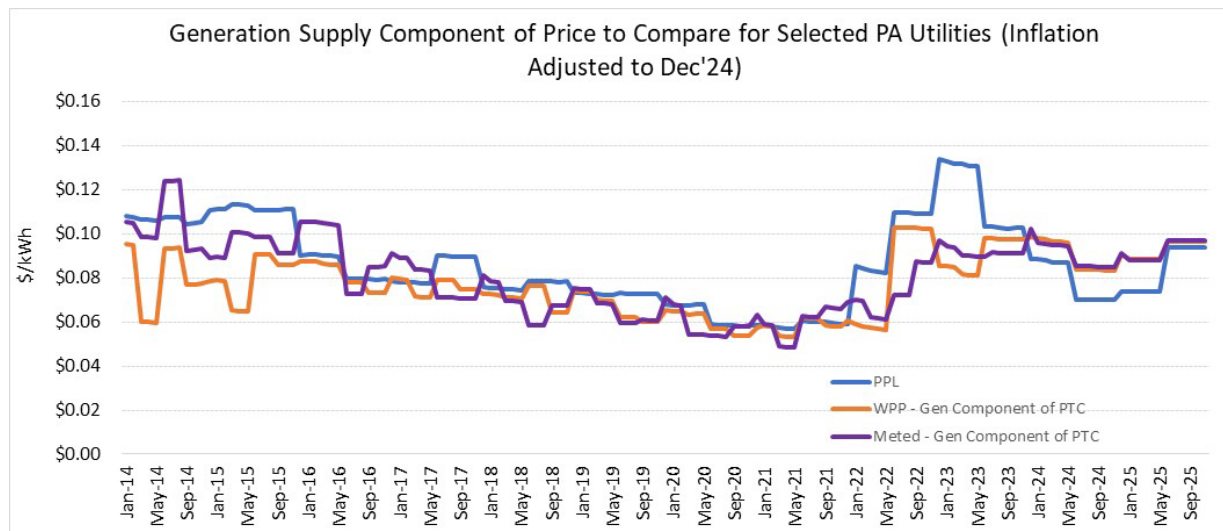


On an inflation adjusted basis, 2025 is not an outlier in terms of generation energy and capacity costs relative to prior years in the study period. Standard offer prices for June 2025 onward were estimated based on known default service procurements and calculated capacity charges. While the final default service rates may still be pending in some utilities, the capacity component is known based on the average residential generation capacity obligation (tag) and the final zonal capacity prices resulting from the completion of the third incremental auction. The differences in monthly generation costs across Pennsylvania utilities are attributable to average monthly customer usage and geographic differences in PJM zonal capacity prices.

For Pennsylvanians, total generation costs as a percentage of their total bill and in inflation adjusted dollar terms are consistent with historical averages for residential customers. In June 2025, capacity prices will rise, but energy prices are currently low relative to recent years and the combined total of energy and capacity costs is consistent with prior years. Other components of the residential bill such as Transmission, Distribution, and Public Policy driven charges have also changed in the past decade as shown in the graph below. Transmission charges, particularly in PPL are the fastest growing components of the bill, especially in inflation adjusted terms.



The Price to Compare (PTC) is the default price for generation supply and is applicable to customers who don't select a competitive supplier. It currently updates semiannually in each utility and is generally a good proxy for generation costs for residential customers. The generation supply component of the PTC is similar to values from approximately three years ago and is comparable to values from approximately ten years ago on an inflation adjusted basis. The PTC for the First Energy utilities shown in the graph below have been adjusted to remove transmission costs which are embedded into the PTC. As the graph shows, generation costs in 2025 are not outliers compared to the inflation adjusted historical record.



Methodology

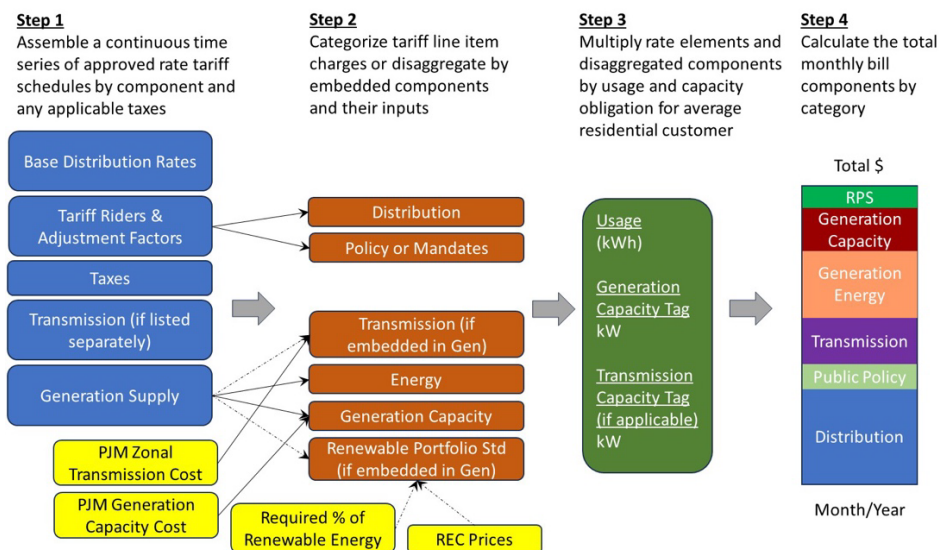
Each utility specific analysis begins with the published tariff rate books approved by the Pennsylvania Public Utility Commission (PAPUC). ETE assembled a continuous dataset of approved rates using its internal tariff library and supplemented with data from the PAPUC docket system when gaps were identified. The rate buildup files include all rate components and relevant meta-data such as tariff sheets and revision numbers.

In some cases, there are multiple embedded costs included in a tariff rate item. This is true for default generation supply charges where energy, capacity, and Renewable Portfolio Standard (RPS) charges are typically comingled. In some utilities, such as the First Energy operating companies, Transmission is also included in the default generation supply charge. ETE unbundled the generation charge into its components using the methods described below:

- **RPS Charges**
 - ETE obtained the historical annual percentages of energy required in each category of renewable or alternative energy and reported average Renewable Energy Credit (REC) prices. Multiplication of the required percentages and prevailing REC prices provides the unit cost in \$/MWh of RPS compliance.
- **Generation Capacity**
 - ETE assembled a time series of zonal capacity prices from the PJM Reliability Pricing Model (RPM) capacity auctions along with scaling factors such as line losses, forecast pool requirements, and zonal scaling factors. The final zonal capacity prices resulting from last incremental auction, if applicable, were used in this analysis.
 - Residential class average generation capacity obligation in kW (aka, “capacity tags”) were used to calculate the average residential customer cost responsibility for capacity
 - The scaled capacity prices were adjusted to daily \$/kW-day values and multiplied by the average residential capacity tag and number of days in a month to determine the total cost of generation capacity.
- **Transmission (if applicable)**
 - In cases where PJM transmission charges are embedded in supply rates, ETE assembled a time series of historical zonal Network Integration Transmission Service (NITS) charges and converted them to \$/kW-day.
 - ETE obtained the average residential transmission capacity obligation values in kW (aka Transmission tags)
 - The transmission cost responsibility for the average residential customer was determined by multiplying the average residential capacity tags scaled for line losses by the effective daily NITS rate and the days in a month.
- **Generation Energy**

- The total cost of the energy component of the generation charge was found deductively. The monthly cost of RPS compliance, generation capacity, and transmission (if applicable) were subtracted from the total cost of default generation supply and the remainder was ascribed to the energy component.
- **Public Policy Charges**
 - In all Pennsylvania utilities, the Universal Service Rider and Act 129 based riders for energy efficiency and conservation were deemed public policy related.

Figure 3: Illustration of Steps to Replicate Monthly Average Residential Electric Bill



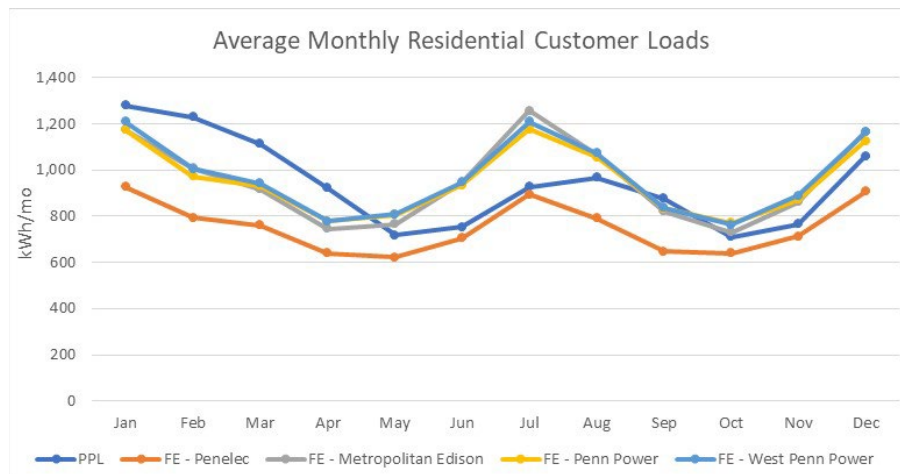
In PA, residential bills were calculated for the following utilities: PPL; Penelec; Metropolitan Edison; Penn Power; West Penn Power; and Duquesne.

In addition to calculating bills in nominal dollars, ETE also calculated an inflation adjusted residential electric bill using the Consumer Price Index for All Urban Consumers (CPI-U) with a reference point of December 2024.

Definition of Average Residential Customer

In Pennsylvania, ETE utilized the data made available to retail energy suppliers participating in the default generation supply service auctions. This data includes total residential customer counts, monthly consumption, and generation capacity obligations for the class. A minimum of 24 months of data was used to develop an annual profile of average residential monthly usage. The average residential consumption patterns by PA utility are shown in the graph below.

Figure 4



This step is important because this analysis is focused on total customer bills. Usage trends vary by utility service area due to climate, average dwelling sizes, and home characteristics such as electric heat. As a result, comparing total bills can be a more relevant assessment of consumer costs than comparing unit costs in \$/kWh, by bill component.

Understanding Your Electric Bill: Maryland

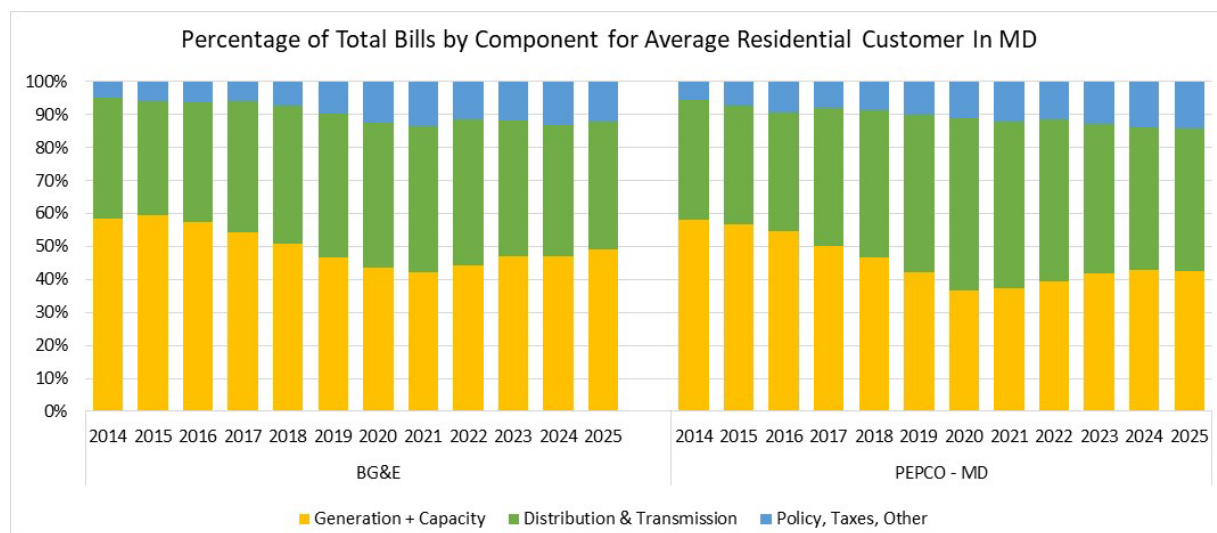
Energy Tariff Experts, LLC (ETE) conducted an analysis of electric utility bills for the average residential customer in the PJM region in order to better understand how costs for different segments of the electric grid have contributed to the total bill over the past decade.

Analysis and Methods

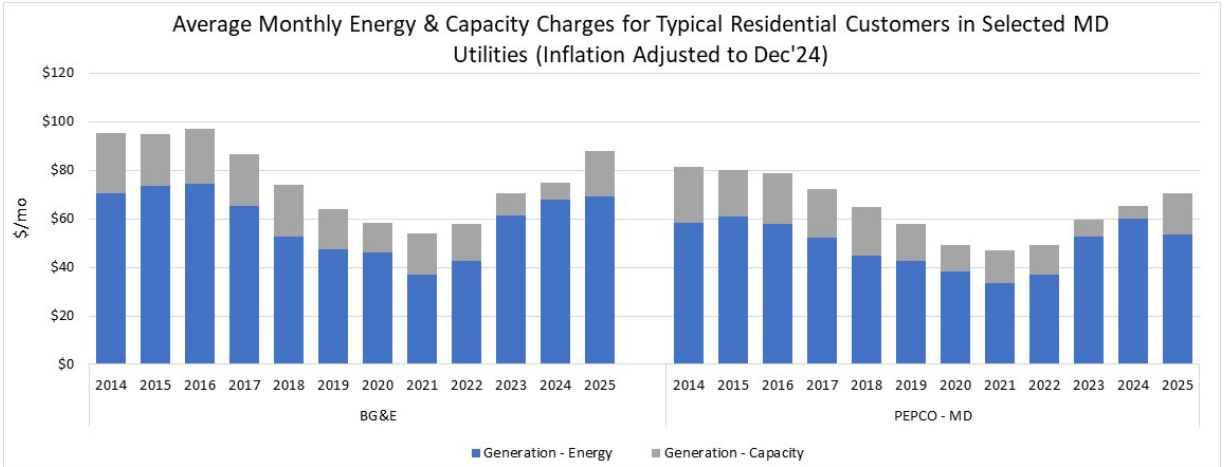
ETE assembled multiple primary sources to replicate a typical residential invoice by component. The electric bill for an average residential customer was replicated based on effective electric tariffs and other relevant data. The end product is additive of the total residential electric costs, by component over the past decade to show growth trends in total bills and underlying cost drivers. To learn more, see the methodology below.

SUMMARY OF FINDINGS

In its analysis of Maryland residential electric bills, ETE found that the generation component of residential customer bills has fluctuated over the study period, averaging approximately 48% of the total bill. In 2025, the residential retail cost of generation as a percentage of customer bills has not increased materially and is consistent with historical averages. Transmission and public policy related charges have grown in the last decade and consume a greater share of the total residential bill. The chart below shows how these impacts change by specific utility:

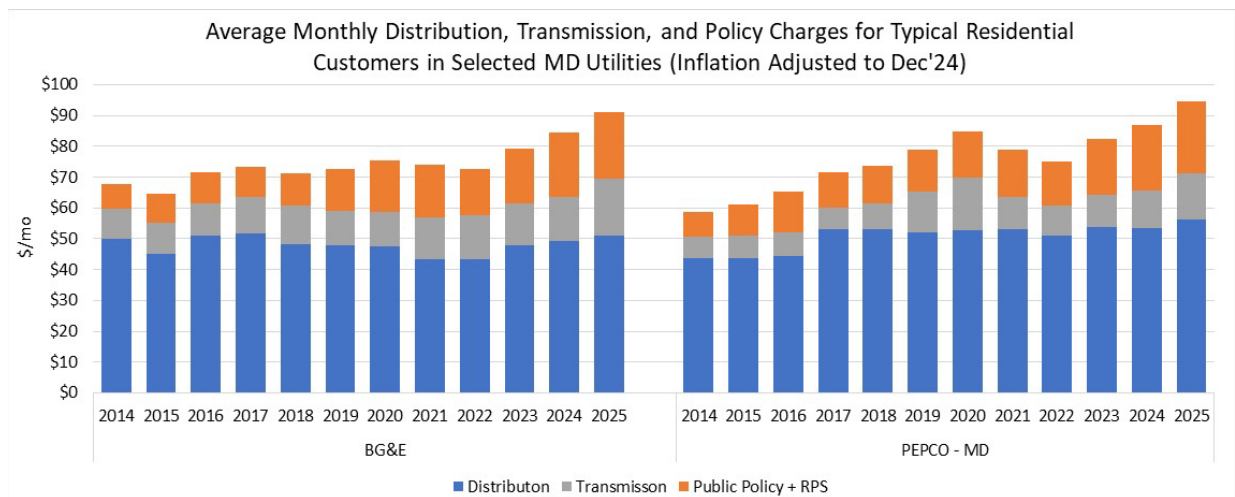


The generation component of the bill consists of both energy and capacity since these markets work together to ensure reliable energy supply. Throughout the study period, costs for energy and capacity have varied, but often in divergent directions.

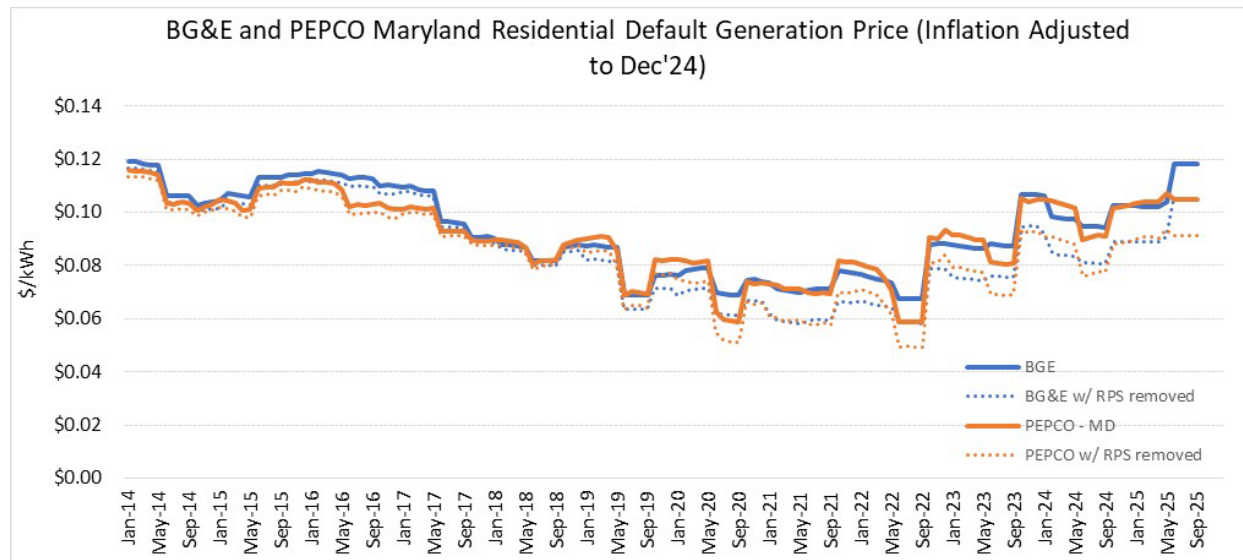


On an inflation adjusted basis, 2025 is not an outlier in terms of generation energy and capacity costs relative to prior years in the study period. Standard Offer generation prices for June 2025 are now publicly posted in Maryland, and the capacity component can be calculated based on the average residential generation capacity obligation (tag) and the final zonal capacity prices resulting from the completion of the third incremental auction. The differences in monthly generation costs across Maryland utilities can be attributable to average monthly customer usage and geographic differences in PJM zonal capacity prices.

For Maryland residents, total generation costs as a percentage of their total bill and in inflation adjusted dollar terms have not increased materially relative to historical averages for residential customers. In June 2025, capacity prices will rise, but energy prices are currently lower relative to recent years, and the combined total of energy and capacity costs is consistent with prior years. Other components of the residential bill such as Transmission, Distribution, and Public Policy driven charges have also changed in the past decade, with most of the growth stemming from increases in transmission and public policy related charges, as shown in the graph below.



The Price to Compare (PTC) is the default price for generation supply and is applicable to customers who don't select a competitive supplier. It updates several times a year and is generally a good proxy for generation costs for residential customers. In unit cost terms, the generation component of the PTC is similar to several other years within the past decade on an inflation adjusted basis. This is shown in the graphic below with generation prices inclusive of the Energy Cost Adjustment and Procurement Cost Adjustment for BG&E and PEPCO-MD, respectively. The light dashed line illustrates the impact of removing Renewable Portfolio Standard (RPS) compliance costs, which are embedded in the PTC, to show the portion of the PTC that is solely attributable to generation.



Methodology

Each utility specific analysis begins with the published tariff rate books approved by the Maryland Public Service Commission (PSC). ETE assembled a continuous dataset of approved rates using its internal tariff library, utility websites, and the PSC docket system. The rate buildup files include all rate components and relevant meta-data such as tariff sheets and revision numbers.

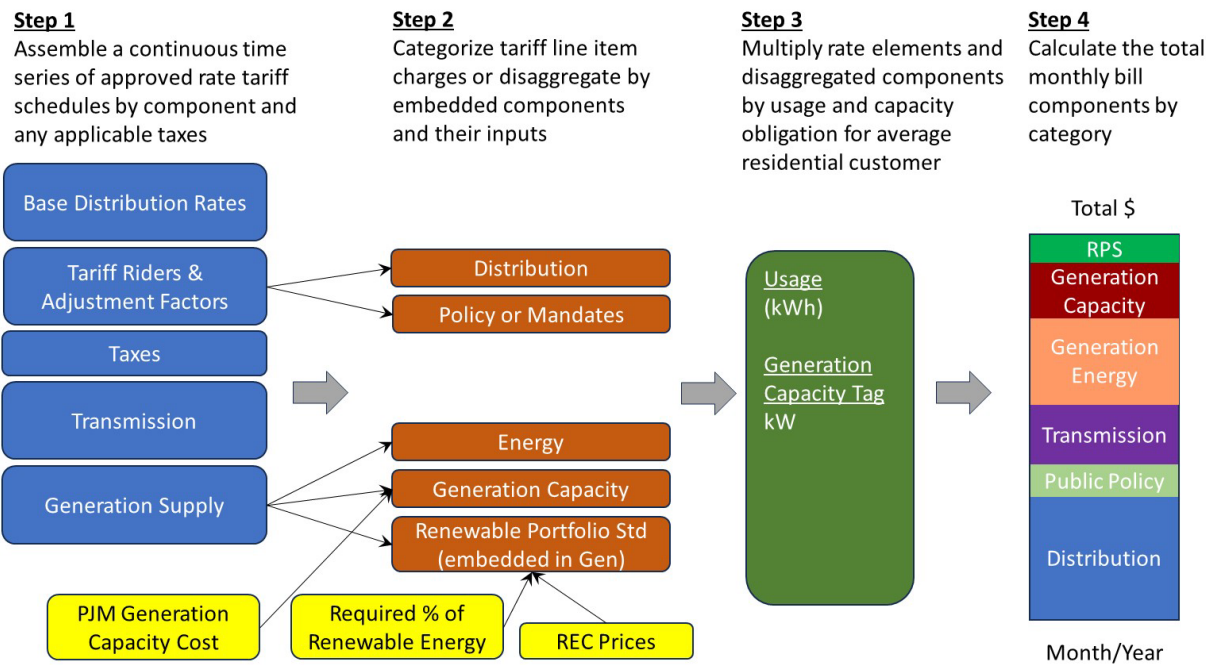
In some cases, there are multiple embedded costs included in a tariff rate item. This is true for standard offer supply charges where energy, capacity, Renewable Portfolio Standard (RPS) charges are typically comingled. ETE unbundled the generation charge into its components using the methods described below:

- **RPS Charges**
 - ETE obtained the historical annual percentages of energy required in each category of renewables and reported average Renewable Energy Credit (REC) prices contained in the annual RPS compliance reports available from the MD PSC. For more recent periods where MD PSC RPS reports are not yet available, ETE multiplied the required percentages of renewable energy and prevailing REC prices to calculate the unit cost in \$/MWh of RPS compliance.
- **Generation Capacity**
 - ETE assembled a time series of zonal capacity prices from the PJM Reliability Pricing Model (RPM) capacity auctions along with scaling factors such as line losses, forecast pool requirements, and zonal scaling factors. The final zonal capacity prices resulting from last incremental auction, if applicable, were used in this analysis.
 - Residential class average generation capacity obligation in kW (aka, “capacity tags”) were used to calculate the average residential customer cost responsibility for capacity
 - The scaled capacity prices were adjusted to daily \$/kW-day values and multiplied by the average residential capacity tag and number of days in a month to determine the total cost of generation capacity.
- **Generation Energy**
 - The total cost of the energy component of the generation charge was found deductively. The monthly cost of RPS compliance and generation capacity were subtracted from the total cost of default generation supply and the remainder was ascribed to the energy component.
- **Public Policy Charges**
 - The riders included in the EmPower Maryland billing line item, the Demand Resources Rider, and Universal Service Fund were designated as public policy related charges in both BG&E and PEPCO-MD. In PEPCO-MD, the Economic Relief & Recovery Rider (Rider ERR) was also labeled as public policy.

- **Taxes**

- The State Electric Environmental Surcharge and Maryland Franchise Tax were identified as taxes. This analysis does not include local taxes such as Montgomery or Prince Georges County taxes that may be applicable.

Figure 5: Illustration of Steps to Replicate Monthly Average Residential Electric Bill



In Maryland, residential bills were calculated for BG&E and PEPCO.

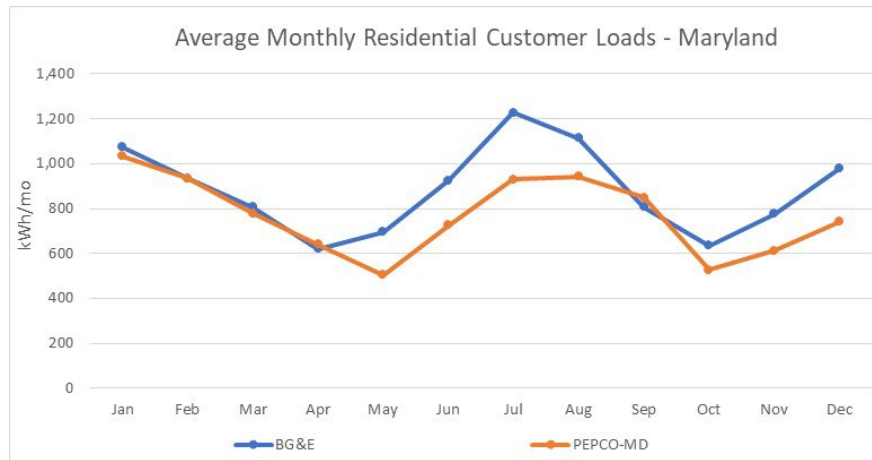
In addition to calculating bills in nominal dollars, ETE also calculated an inflation adjusted residential electric bill using the Consumer Price Index for All Urban Consumers (CPI-U) with a reference point of December 2024.

Definition of Average Residential Customer

In Maryland, ETE utilized a combination of the data made available to retail energy suppliers, and rate case workpapers including test year sales and sales forecasts to determine the monthly usage for the average residential customer. This data includes total residential customer counts, monthly consumption, and generation capacity obligations for the class. In BG&E, ETE obtained usage data for approximately 36 months. In PEPCO, ETE relied upon the sales forecast submitted to the PSC in its application for a Multi-Year Rate Plan. The analysis is focused on the default residential distribution rate in PEPCO and excludes the Time of Use (TOU) option. ETE analysis shows that although approximately ten percent of PEPCO customers are on the residential TOU rate option, their average monthly usage is 40% greater than the default residential distribution rate option and their inclusion in the dataset could skew the monthly averages. As a result, ETE selected the default residential rate customer population to

approximate the average residential customer. The average residential consumption patterns by Maryland utility are shown in the graph below.

Figure 6



This step is important because this analysis is focused on total customer bills. Usage trends vary by utility service area due to climate, average dwelling sizes, and home characteristics such as electric heat. As a result, comparing total bills can be a more relevant assessment of consumer costs than comparing unit costs in \$/kWh, by bill component.

Understanding Your Electric Bill: Ohio

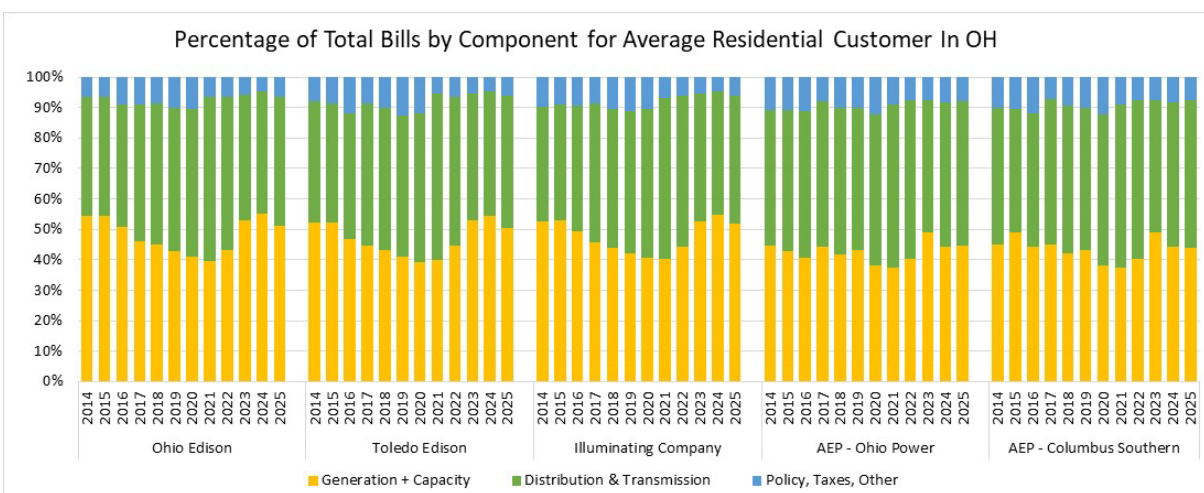
Energy Tariff Experts, LLC (ETE) conducted an analysis of electric utility bills for the average residential customer in the PJM region in order to better understand how costs for different segments of the electric grid have contributed to the total bill over the past decade.

Analysis and Methods

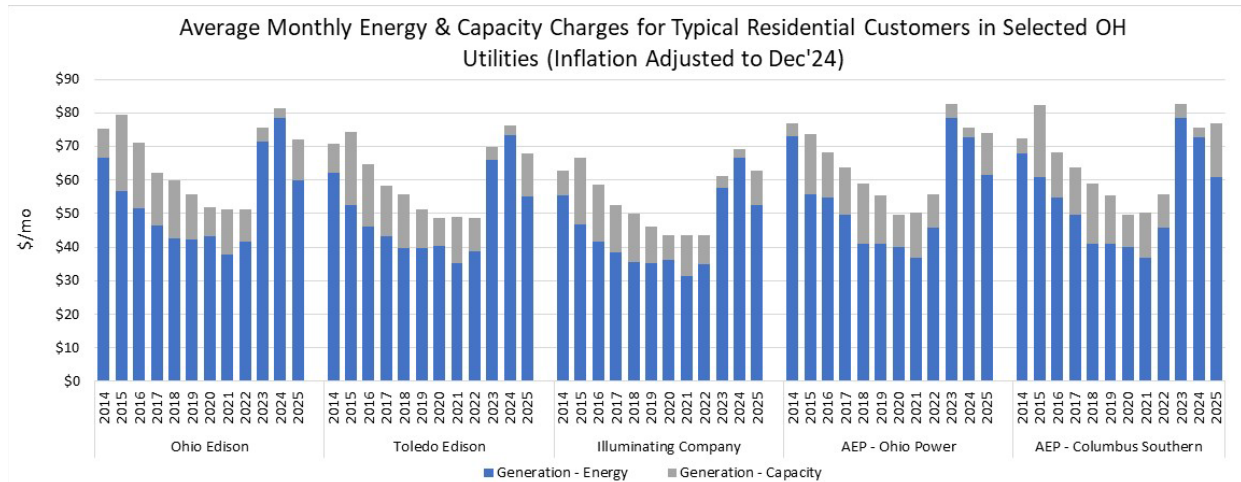
ETE assembled multiple primary sources to replicate a typical residential invoice by component. The electric bill for an average residential customer was replicated based on effective electric tariffs and other relevant data. The end product is additive of the total residential electric costs, by component over the past decade to show growth trends in total bills and underlying cost drivers. To learn more about analysis and methods, see the attached methodology.

SUMMARY OF FINDINGS

In its analysis of Ohio residential electric bills, ETE found that the generation component of residential customer bills has fluctuated over the study period, averaging approximately 46% of the total bill. In 2025, the residential retail cost of generation as a percentage of customer bills has not increased materially over 2024 levels and is consistent with historical averages. The data indicates that transmission charges have been growing in the Ohio utilities studied, but some of this growth has been offset by declining distribution costs, in inflation adjusted terms. The chart below shows how these impacts change by specific utility:

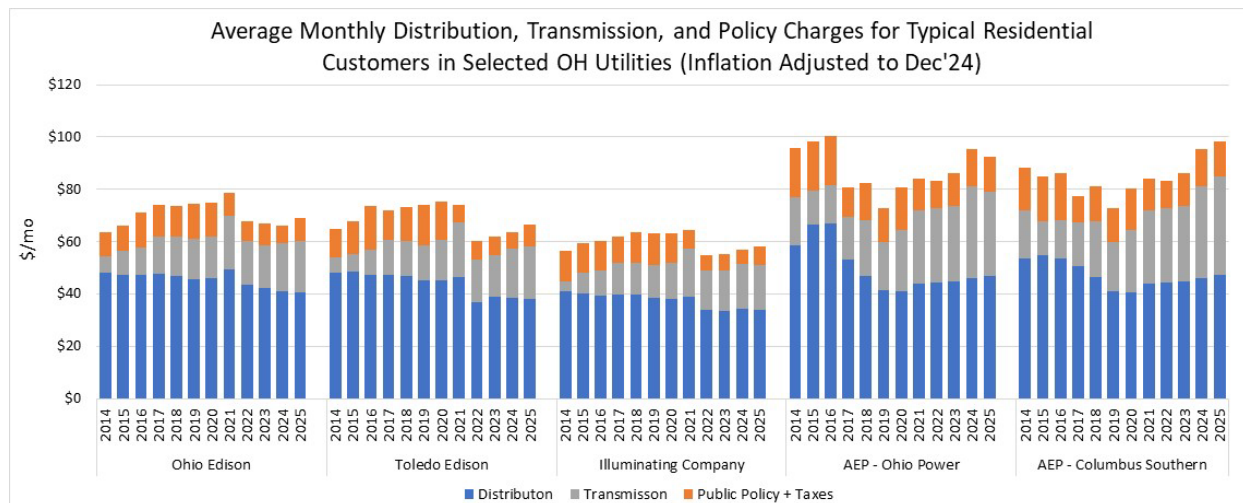


The generation component of the bill consists of both energy and capacity since these markets work together to ensure reliable energy supply. Throughout the study period, costs for energy and capacity have varied, but often in divergent directions.

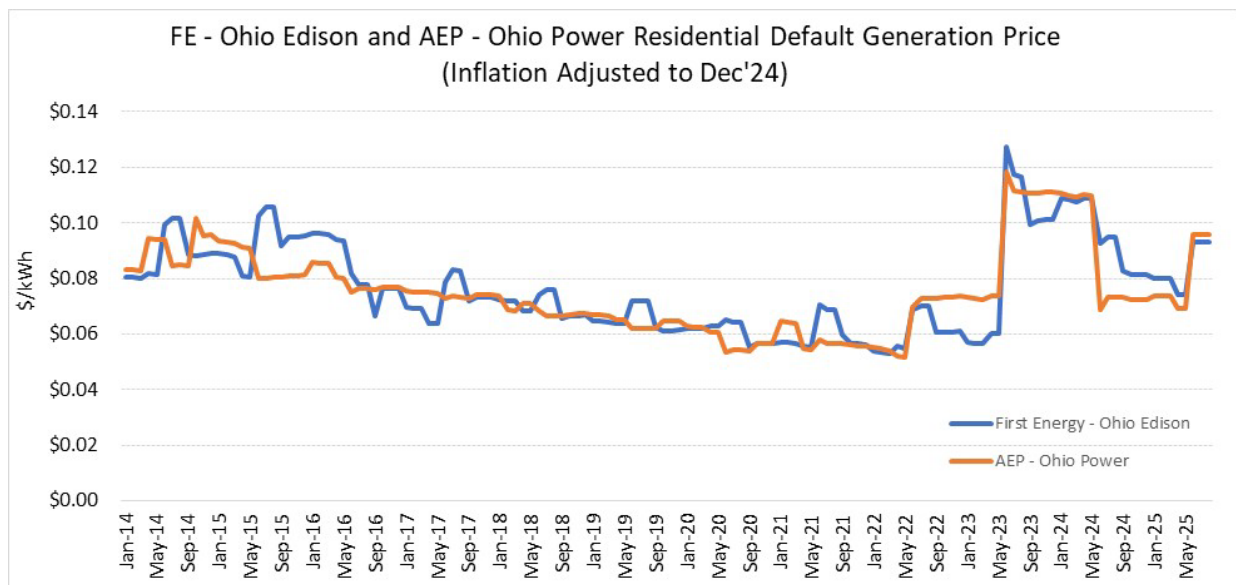


On an inflation adjusted basis, 2025 is not an outlier in terms of generation energy and capacity costs relative to prior years in the study period. Default service prices for June 2025 onward were estimated based on known default service procurements and calculated capacity charges. While the final default service rates may still be pending in some utilities, ETE has replicated utility workpapers used to determine the capacity component based on the final zonal capacity prices resulting from the completion of the third incremental auction. The differences in monthly generation costs across Ohio utilities are attributable to average monthly customer usage, geographic differences in PJM zonal capacity prices, and differences in default generation service auction procurement schedules.

For Ohioans, total generation costs as a percentage of their total bill and in inflation adjusted dollar terms are consistent with historical averages for residential customers. In June 2025, capacity prices will rise, but energy prices are currently low relative to recent years, and the combined total of energy and capacity costs is consistent with prior years. Other components of the residential bill such as Transmission, Distribution, and Public Policy driven charges have also changed in the past decade, with transmission growing and increasingly displacing distribution costs in the total monthly billed amount. This trend is shown in the graph below.



The current generation supply charges in AEP – Ohio Power and First Energy’s Ohio Edison service area are actually lower than values experienced in portions of 2023 and 2024. ETE selected the default generation charges, otherwise referred to as the Price to Compare (PTC) because they are posted publicly and remain a good proxy for generation costs for customers who chose a retail supplier or participate in a Community Choice Aggregation (CCA). In Ohio, the PTC updates annually and has a summer and winter season. On an inflation adjusted basis, generation supply charges are comparable to values from a decade ago. The charges shown for First Energy are inclusive of energy, capacity, reconciliation riders, and generation uncollectibles. For AEP, the current generation PTC tariff structure dates from 2015 and includes energy, capacity, and reconciliation riders. Although two utilities are shown in the graph below, the patterns for First Energy and AEP’s other Ohio service territories are similar.

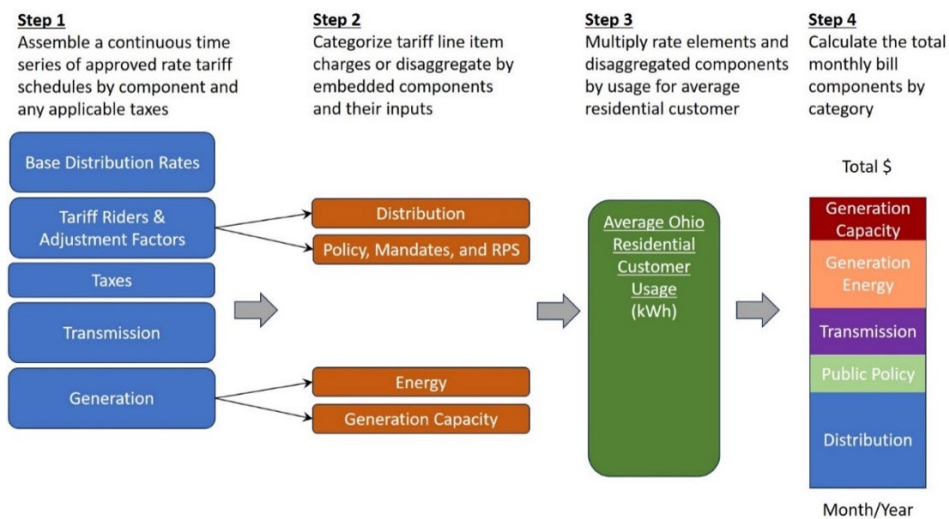


Methodology

Each utility-specific analysis begins with the published tariff rate books approved by the Public Utility Commission of Ohio (PUCO). ETE assembled a continuous dataset of approved rates using its internal tariff library and supplemented with data from the PUCO docket system when gaps were identified. The rate buildup files include all rate components and relevant meta-data such as tariff sheets and revision numbers.

In the major utilities in Ohio, tariffed rates for generation supply are generally bifurcated into energy and capacity. The generation component for the Ohio Renewable Portfolio Standard (RPS) is also billed as a separate rider and is categorized as a policy charge. The process used to calculate a disaggregated residential electric bill is shown in Figure 5 below.

Figure 7: Illustration of Steps to Replicate Monthly Average Residential Electric Bill



In Ohio, residential bills were calculated for the following utilities: FE – Ohio Edison, FE – Illuminating Company; FE – Toledo Edison; AEP – Ohio Power; and AEP – Columbus Southern.

The table shows the riders that were labeled as being public policy for each utility.

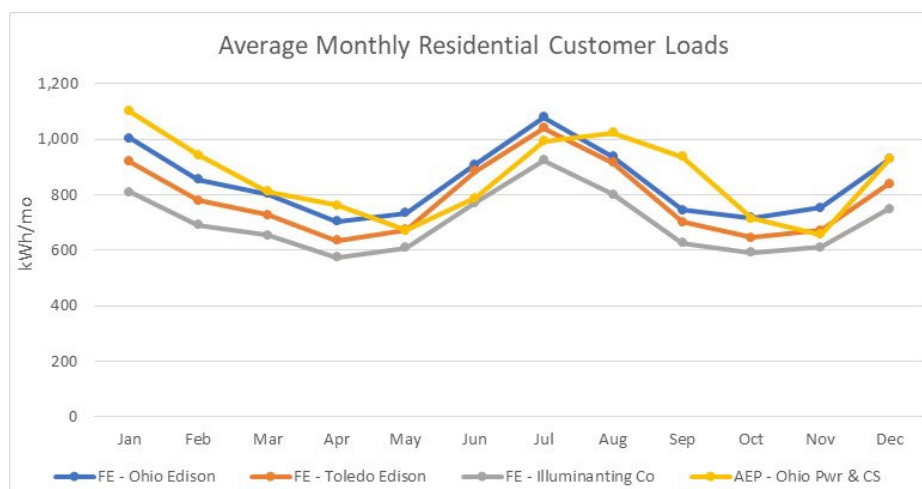
First Energy Companies	American Electric Power
Universal Service Fund	Universal Service Fund
Demand Side Management and Energy Efficiency; Conservation Support Rider	Energy Efficiency and Peak Demand Reduction Cost Recovery and its successor
Alternative Energy Resource Rider	Alternative Energy Rider
Legacy Generation Resource	Legacy Generation Resource
Solar Generation Fund	Solar Generation Fund
Energy Efficiency Cost Recovery	Smart City Rider
	Ohio First Rider

In addition to calculating bills in nominal dollars, ETE also calculated an inflation adjusted residential electric bill using the Consumer Price Index for All Urban Consumers (CPI-U) with a reference point of December 2024.

Definition of Average Residential Customer

In Ohio, ETE utilized the load data made public to retail energy suppliers participating in the default generation supply service auctions and customer shopping statistics for the competitive market. This data was cross referenced with workpapers from utility filings at PUCO. Data used to estimate the average residential customer in each utility includes customer counts and monthly consumption for the class. A minimum of 24 months of data was used to develop an annual profile of average residential monthly usage. The average residential consumption patterns by Ohio utility analyzed is shown in the graph below.

Figure 8



This step is important because this analysis is focused on total customer bills in total dollars. Usage trends vary by utility service area due to climate, average dwelling sizes, and home characteristics such as electric heat. As a result, comparing total bills can be a more relevant assessment of consumer costs than comparing unit costs in \$/kWh, by bill component.